

Digital Multiverses: A Computational Framework for Counterfactual Storyboarding in Action Cinema

George Wu

Based on the Research by Elaine Chu, Andrew Liao, and George Wu

June 9, 2026

Abstract

This report details a generative AI pipeline designed to explore “what-if” (counterfactual) scenarios in action cinema. While traditional script development relies on manual iteration to explore narrative pivots, our system leverages a synchronized coupling of fine-tuned Large Language Models (LLMs) and Diffusion-based image synthesis to automate this process. Specifically, we employ Gemma3 models adapted via QLoRA to ensure narrative causality and Stable Diffusion v1.5 with DreamBooth stylization to produce consistent, non-photorealistic visual storyboards. A critical component of our methodology is the strategic fine-tuning of both language and image models, allowing for the preservation of character essence and thematic integrity across divergent plot trajectories. We demonstrate the system’s efficacy through case studies—notably a moral-reversal arc in *Spider-Man* and a mechanical-failure pivot in *Speed*—showing that the framework can generate both logically sound scripts and stylistically cohesive storyboard panels. Ultimately, this work provides screenwriters, directors, and educators with a “digital sandbox” for pre-visualization, deepening narrative analysis and offering a scalable tool for creative inspiration during the high-stakes pre-production phase of film development.

1 Introduction

In the craft of screenwriting, a narrative is essentially a sequence of causal dependencies. Every major plot point acts as a logical gate; if a character makes a different choice, the entire structural integrity of the story shifts. This project explores the intersection of narrative theory and generative AI through the lens of *Counterfactual Analysis*—the study of “what-if” scenarios that deviate from established film history. For directors and scriptwriters, the ability

to visualize these alternate realities is often a manual, labor-intensive process involving extensive rewriting and re-storyboarding. Our work introduces a computational framework that automates this exploration. By providing a single counterfactual trigger—such as changing a pivotal survival or a mechanical failure—our system generates a logically consistent alternate script and a corresponding visual storyboard. The traditional pre-production workflow often suffers from "narrative inertia," where the sheer volume of work required to explore a new direction discourages creative experimentation. Our framework seeks to eliminate this friction by treating the screenplay not as a static document, but as a dynamic system of variables. By leveraging a synchronized coupling of fine-tuned Large Language Models and Diffusion-based image synthesis, we allow creators to iterate at the speed of thought. This transition from manual drafting to algorithmic generation enables a "Digital Multiverse" approach, where dozens of narrative branches can be audited for emotional resonance and logical soundness before a single cent is spent on physical production.

1.1 Narrative Logic and Constraints

Action cinema is particularly suited for this analysis because it often relies on high-stakes, binary outcomes. For instance, in the 1994 film *Speed*, the internal logic is dictated by the bus's velocity. If the velocity drops, the story ends or transforms. Our system treats these plot points as variables in a statistical distribution, allowing us to sample new narrative trajectories that maintain Character consistency, causal coherence, as well as thematic integrity. We want to ensure the characters react to the new stimuli in a way that aligned with their established characteristics and personalities, maintaining the "chain of consequence", meaning each new scene afterward is logically aligned with twist during the plots. Last but not least, preserving the sentiment core of the film despite the counterfactual change.

By bridging the gap between Large Language Models (LLMs) and Diffusion-based image generation, we offer a "digital sandbox" for the pre-production phase, enabling creators to stress-test their scripts and find new avenues of creative inspiration.

2 Background and terminology

To facilitate a shared understanding between the technical and creative domains, this section defines the core concepts and computational tools utilized in our pipeline.

2.1 Counterfactual Narratives

In narrative theory, a *counterfactual* is a conditional statement that explores what would have happened if a specific past event had been different. In the context of this framework, we define a **Counterfactual Trigger** as the precise moment of divergence from the original script. From a statistical perspective, this is a perturbation of the input data that forces the model to recalculate the probability of subsequent plot points.

2.2 Large Language Models (LLMs) and Fine-Tuning

The “brain” of our system is the **Gemma3** model. To better understand, we need to know what LLM is in general. Basically, it is a mathematical model trained on vast corpora of text to predict the next logical word in a sequence. Therefore, in our counterfactual scenario, we are essentially “weighting” certain outcomes. For example, if Uncle Ben survives, the probability of the word “responsibility” appearing in Peter’s dialogue might drop, while “profit” or “fame” increases. For writers, think of this as an extremely well-read assistant who has analyzed thousands of screenplays. While we treat standard models as generalists, we use *Quantized Low-Rank Adaptation* (QLoRA) to specialize the model. This allows us to keep the “foundational knowledge” of the model intact while teaching it the specific formatting, pacing, and tropes of action cinema without the need for massive computing power.

2.3 Diffusion Models and Storyboarding

For visual generation, we employ **Stable Diffusion v1.5**. This is a pre-trained model that is particularly great for our task specific project. It is a process where the model starts with a field of random static (noise) and gradually shapes it into an image based on a text prompt. This is analogous to a concept artist sketching rough shapes before refining the details. Since the inputs that we fed into the model are all in sketch style/storyboard style, we used DreamBooth to fine-tune it. These are techniques used to teach the model a specific visual “style.” In our project, these were used to ensure every output looks like a hand-drawn **Sketch Pencil Storyboard**, maintaining aesthetic consistency across different generated scenes.

2.4 Evaluation Metrics: Loss and Convergence

In data science, we measure a model’s success through a **Loss Function**. It’s basically a numerical value representing the difference between the model’s output and the desired result.

Our goal is to “minimize loss” so that the model can best replicate the task we were asking for. Moreover, the term **convergence** is also widely used in this setting. It means the point where the model has learned as much as it can from the data. And the model is ready to do inference for our purposes. For a director, a model that has not converged is like an unrehearsed actor—it understands the lines but lacks the correct delivery. Nevertheless, to ensure that the model does not forget general objects like cars or busses while focusing on the sketch style, we utilize a Prior Preservation strategy, which acts as a structural reference library for the AI.

Noticeably, even though our goal is to minimize the loss, that doesn’t necessarily means the model is doing better than its counter part(without fine-tuning). We only fine-tune it so that the model can learn how to mimic the prompt that we fed to them and tried to recreate based on that. It still requires human evaluation to decide whether the generated output is better or worse. Nonetheless, when we train these models, we are navigating the **Latent Space**—a mathematical map of all possible images and words. The model learns the “shape” of an action movie. When we introduce a counterfactual, we are essentially asking the model to find a new path on that map that still stays within the boundaries of the “Action Movie” territory. A critical setting within these language models is Temperature, a hyperparameter that governs the randomness of the output. A low temperature produces more literal, predictable results, whereas a higher temperature encourages the model to take creative risks by exploring less probable connections within the Latent Space.

3 Methodology

In order to create logically sound alternate movie plots, we utilized Gemma3(4B-it), a large language model (LLM) selected for its strong reasoning and instruction-following foundations. While standard models can tell stories, they often lack the logical consistency required for action movie scripts. The Large Language model is the model that “reads” through a large amount of information on the internet and instead of memorizing this information, LLM will learn the patterns of words and treat words like coordinates on a map, where words with similar meanings are grouped close together.

We employed a technique called QLoRA (Quantized Low-Rank Adaptation) for efficient model adaptation. This is one of the methods for fine-tuning the model, which basically says we want to train the model so it understands the specific task we are trying to do but also

remember the knowledge that it has learned before. So the way QLoRA works is it updates a small fraction of the model’s weights using ”adapters” instead of updating the entire model’s parameters, which is computationally expensive.

To ensure high-quality script generation, we monitored performance using WandB(Weights and Biases). We can view them as script supervisors or like technical monitors that record every single take to see what is working and what isn’t. It will record how well the model can write the scripts that we want and minimize the loss (basically saying how many mistakes the model made). We also use WandB Random Hyperband Search to automatically test dozens of different ”settings” at once to find the best combination for generating coherent movie plots. Once the text is generated, the system ”draws” the scenes using Stable Diffusion.

To ensure a consistent ”hand-drawn” aesthetic across the storyboard, we used DreamBooth. This procedure takes a small set of images (8–10) in a specific style—in our case, pencil-sketch storyboards—and teaches the model to replicate that look. One good thing about this model is that we don’t necessarily need a lot of inputs(images) to fine-tune it. Notably, the way we fine-tune the diffusion model is slightly different from the normal model. During the process, the model will add noises by itself to the images and compare the actual noises versus predicted noises(MSE). This means that the model is not generating the output(image) during training, but instead predicting the noises that help us to recreate the scene or photo. Therefore, if the actual noises are extremely different from the predicted noises, which means we have high error, the model adjusts its internal ”brain” (weights) to be more accurate next time.

The images we fed to the model are all in either sketch style or comic style. Essentially, we are telling the model that ”When I use this trigger word, ignore the photorealism you learned from billions of internet photos and only look at this specific aesthetic.” One key difference is that we purposely ”overfitted” the model slightly. While typically avoided in data science, this ensured the images looked like sketches rather than realistic photos. During our experiment we saw that a lot of time DreamBooth will produce real movie scenes or images, which is not ideal for us. Another feature in DreamBooth is Prior preservation, which we use to prevent the model from forgetting general objects (like cars or buses) while it focuses on the sketch style.

4 Experimental Results and Case Studies

To validate the efficacy of our framework, we conducted several experiments using iconic action film scripts. The most prominent case study involved a counterfactual intervention in the film called *spider-man*.

4.1 Case Study: The Ethical Divergence in *Spider-Man*

To test the framework’s ability to handle character-driven drama, we applied a counterfactual to the 2002 film *Spider-Man*. The prompt removed the foundational trauma of the protagonist:

“What if Uncle Ben had not been killed?”

The model correctly identified that without the guilt of Uncle Ben’s death, the famous quote “With Great Power comes Great Responsibility” will never happened. As a result, the generated script follows a trajectory of **Moral Compromise**.

Unlike a standard hero’s journey, the LLM generated a narrative where Peter Parker uses his abilities for *personal gain* and *popularity* (winning fights and performing for money). The conflict with the antagonist (Green Goblin) is motivated by a *personal vendetta* after Harry is injured, rather than a sense of civic duty. The resolution is categorized by **brutality**, leading to a thematic shift in which the protagonist becomes a figure of fear even for his own family.

The fine-tuned Gemma3 model successfully flipped the script from a high-octane chase to a high-stakes forensic thriller. As shown in Table 1, the model did not simply end the story with the explosion; it generated a logical second act focused on the aftermath and the search for the antagonist, Howard Payne.

Table 1: Spider-Man: Heroic vs. Hubristic Trajectories

Milestone	Original (Hero)	Counterfactual (Hubris)
Catalyst	Uncle Ben’s death creates a moral vowing.	Uncle Ben survives; Peter remains self-centered.
Motivation	Altruistic responsibility.	Fame, fortune, and later, revenge.
Climax	Rescue of civilians; rejection of MJ to protect her.	Brutal defeat of Goblin; alienation of Ben and May.

4.2 Visual Storyboarding Performance



(a) LR with $7e^6$ with small training step



(b) LR with $1e^5$ with large training step

Figure 1: Example storyboards with different settings of hyperparameter

Noticeably, just a slight change in the hyperparameter can heavily affect the way storyboard is generated. On the left, the image has more focus on the facial details and how the characters are shaped. On the right the image is less detailed but the model is trying to generate the conversation, even though there’s no actual words in the blocks. Often time we need human evaluation to decide which aspects of the storyboards are desired for our purposes. From our experiments, we notice that parameters are super sensitive and therefore it’s really hard to find the best set of hyper parameters that generates the ”perfect” storyboards.

5 Conclusion and Shortcomings

The convergence of generative artificial intelligence and narrative theory marks a significant shift in the pre-production landscape. This framework demonstrates that computational models, when properly tuned, can act as more than mere mirrors of existing content; they can serve as structural analysts capable of identifying the ”genetic code” of a screenplay. By automating the visual and narrative consequences of a single script change—whether it be the moral decay of a hero in *Spider-Man* or the explosive failure of a mechanical constraint in *Speed*—we provide filmmakers with a high-fidelity ”what-if” engine. This allows for the stress-testing of story beats and the exploration of character arcs that might otherwise remain undiscovered in the traditional writing process.

However, the transition from algorithmic output to a director’s vision is not a replacement

for human intuition, but an augmentation of it. The current technical shortcomings are the sensitivity of hyperparameters and the lack of objective narrative metrics to evaluate the outputs, which requires many different perspectives to achieve. In the same way that a director collaborates with a Director of Photography to find the right exposure, or a writer works with an editor to find the right pacing, this AI framework requires a creative lead to steer its outputs toward emotional resonance. The data provides the structural logic, but the artist provide the soul. None the less, like we mentioned earlier, our model frameworks are super sensitive to parameter settings. Therefore, even we are able to say the model fulfill the tasks we are given to them, it doesn't necessary mean it has reach the goal we want.

Looking forward, the goal is to transcend the "Action Movie" specialist role of current models. We envision a future where this system can handle the subtle subtext of character-driven dramas and the complex world-building of speculative fiction with equal dexterity. By refining our optimization methods to reflect dramatic tension rather than just statistical probability, we aim to build a truly collaborative partner for the development room—a digital sandbox where every creative pivot can be visualized, analyzed, and refined before the first frame of principal photography is ever shot.

A Technical Implementation Details

This appendix provides the specific "production specs" for the models used in the framework, intended for readers interested in the statistical calibration of the narrative engine.

The Github Link provided here that you can run the model and do some experiements with youself: Github Repository

A.1 Language Model Configuration (Gemma3)

To achieve the logical consistency seen in the *Spider-Man* and *Speed* case studies, we utilized **QLoRA** for parameter-efficient fine-tuning. The training environment was configured as follows:

- **Quantization:** 4-bit NormalFloat (NF4) to maximize memory efficiency without sacrificing narrative nuance.
- **Rank (r):** 16, with an Alpha (α) of 32, allowing the model to adapt its internal weights to script-specific formatting.

- **Optimizer:** Paged AdamW 8-bit, chosen for its stability during the non-linear optimization of complex plot structures.

QLORA functions by updating a small fraction of weights using "adapters" : $W_{updated} = W_{frozen} + (A \times B)$. This shows that the "Foundational Knowledge" (W_{frozen}) stays the same while only a small "Adapter" ($A \times B$) is trained.

A.2 Diffusion Model Hyperparameters

The visual style of the storyboards was standardized through iterative testing of Learning Rates (LR) and training steps. As discussed in the main text, the model is highly sensitive to these values.

Table 2: Visual Model Training Iterations

Learning Rate	Steps	Resulting Aesthetic
1×10^{-5}	1000	Optimal: Balanced sketch detail.
7×10^{-6}	500	Underfit: Retains too much photo-realism.
1×10^{-6}	1000	Failed Convergence: Blurred compositions.

A.3 Prior Preservation Strategy

To prevent the model from "forgetting" how to draw basic human forms while learning the "Sketch Pencil" style, we utilized a **Prior Preservation Dataset**. This acts like a reference library for the AI, ensuring that while it adopts a new artistic style, it maintains the structural anatomy required for cinematic storyboarding.

B Prompt Engineering for Directors

The quality of the "What-if" output is heavily dependent on the input prompt. We found that the most effective prompts followed a **Trigger** $\rightarrow$ **Consequence** structure.

1. **The Trigger:** A specific deviation from the original script (e.g., "The thief escapes without Ben's death").
2. **The Constraint:** A stylistic instruction for the visual engine (e.g., "hand-drawn storyboard, 3 panels, expressive faces").

C Glossary of Data Science for Filmmakers

- **Inference:** The act of the AI "performing" or generating a script based on your prompt.
- **Latent Space:** The multi-dimensional "map" of ideas where the AI finds the connections between your characters and new plot points.
- **Noise:** The random static the image model starts with before it "focuses" into a storyboard.